



UNITED INTERNATIONAL UNIVERSITY
Department of Computer Science and Engineering (CSE)
Course Syllabus

Part A: Introduction

1	Course Title	Data Structure and Algorithms - I
2	Course Code	CSE 2215
3	Pre-requisites	CSE 1111
4	Course Type	Core Course
5	Credit Hours	3.00
6	Section	S
7	Semester	Spring2026
8	Class Hour	Sun-Wed 11:11 AM to 12:30 PM Room 305
9	Course Instructor's Information	Ashraful Islam Paran(AIP) paran@cse.uiu.ac.bd Office: 319(A)
	Counselling Hours	Check ELMS
10	Course Rationale	This course has been designed to provide a solid foundation about the data structure and algorithms used in computer science. This course will give insights about the pros and cons of different data structures and algorithms.
11	Course Objectives	The objectives of this course are: • To familiarize the basic data structures (array, linked list). • To familiarize complex data structures (queue, stack, priority queue) using basic data structures • To use suitable data structures for different algorithms • To introduce the algorithms and their complexity and use cases

Part B: Content of the Course

12	Course Contents (approved by UGC)	Complexity Analysis: Big Oh, Big Omega, Lower and Upper Limit, Best and Worst Cases, Recursion, Arrays: Memory Mapping, Vector Implementation, Sorting, Linear Time Sort, Insertion Sort, Selection Sort, Linked Lists: Single Linked List, Double Linked List, Circular Linked List, Stack: Implementation using Array and Linked Lists, Application, Tower of Hanoi, Postfix Expression, Queue: Introduction to Queue, Implementation of Queue using Arrays and Linked Lists, Applications, General Tree: Introduction, Definition, Binary Tree: Introduction, Implementation using Array and Pointers, Tree Traversal Algorithms, Binary Search Tree: Operations, Insertion, Deletion, Properties, Tree Applications, Heap and Priority Queue: Introduction, Heap Sort and Application of Priority Queue, Graph Representation: Implementation using Adjacency Matrix and Adjacency List, Graphs: BFS and DFS using Adjacency Matrix and Adjacency Lists, Application of Graphs, Search Algorithms, Set Operations: Make set, Find, Union.		
13	Course	COs	Description	

Outcomes (COs)	C01	Able to choose appropriate data structure as applied to specified problem definition.
	C02	Able to handle operations like searching, insertion, deletion, traversing mechanism etc. on various data structures.
	C03	Able to use linear and non-linear data structures like stacks, queues, linked list etc.

14	Mapping of COs and Program outcomes												
	COs	Program Outcomes(POs)											
		P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011	P012
	C01			C									
	C02			C									
	C03				C								
15	Mapping COs with Teaching-Learning and Assessment Strategy												
	Class	Topics/Assignments									Course Outcomes (COs)		
	1	Introduction to Data Structures and Algorithms, Getting Started with Insertion Sort									CO1, CO2		
	2	Complexity Analysis: Single Loop, Nested Loop									CO1, CO2		
	3	Complexity Analysis: Insertion Sort (Best, Worst, Average cases), Growth of functions (Big-O, Ω , Θ)									CO1, CO2		
	4	Arrays: Memory Mapping, 1D Array, 2D Array (Row-major, Column-Major)									CO1, CO2		
	5	Introduction to Recursion									CO2, CO3		
	6	Arrays: Linear and Binary Search (Best, Worst, Average cases)									CO2		
	7	Arrays: Selection Sort, Linear Time Sort (Counting Sort)									CO1, CO2		
	8	Linked List: Singly (Creation, Display, Insertion, Deletion)									CO2, CO3		
		9	Linked List: Doubly, Circular (Creation, Display, Insertion, Deletion)									CO2, CO3	
10		Stack: Implementation using Arrays and Linked Lists									CO1, CO2, CO3		
11		Queue: Implementation using Arrays and Linked Lists									CO1, CO2, CO3		
	12	Review											

MID EXAM		
13	Stack: Application, Tower of Hanoi, Infix to Postfix, Postfix Expression Evaluation	C02, C03
14	Introduction to Trees, Properties of Trees	C02, C03
15	Binary Tree: Implementation using Arrays and Pointers	C02, C03
16	Tree Traversal Algorithms, Tree Applications	C02, C03
17	Binary Search Tree: Creation, Insertion, Deletion, Properties	C01, C02, C03
18	Heap and Priority Queue: Introduction	C01
19	Heap Sort and Applications of Priority Queue	C01, C02, C03
20	Introduction to Graphs, Properties of Graphs	C02, C03
21	Graph Representations: Implementation using Adjacency Matrix and Adjacency List	C01, C02, C03
22	Graph Traversals: BFS and DFS using Adjacency Matrix and Adjacency Lists	C02, C03
23	Applications of Graphs, Directed Acyclic Graph, Topological Sort	C02, C03
24	Review	
FINAL EXAM		

Part C: Assessment and Evaluation Methods

Assessment Strategy	Assessment Types	Marks
Formative Assessment	Attendance	5%
	Assignments	5%
	Class Tests	20%
Summative Assessment	Mid Term	30%
	Final Exam	40%

Grading System

Letter Grade	Marks %	Grade Point	Letter Grade	Marks%	Grade Point
A (Plain)	90-100	4.00	C+ (Plus)	70-73	2.33
A- (Minus)	86-89	3.67	C (Plain)	66-69	2.00
B+ (Plus)	82-85	3.33	C- (Minus)	62-65	1.67
B (Plain)	78-81	3.00	D+ (Plus)	58-61	1.33
B- (Minus)	74-77	2.67	D (Plain)	55-57	1.00
			F (Fail)	<55	0.00

Part D: Learning Resources

Text Book	1. Introduction to Algorithms – Thomas H. Cormen (4 th edition, MIT Press & McGraw Hill, 2022) 2. Data Structures – E. M. Reingold (Addison Wesley Publication, 1998)
Reference	1. Data Structure and Algorithms in C++ - Goodrich, Tamassia (2 nd edition, John Wiley and Sons Inc., 2003)

Appendix-1: Program outcomes

POs	Program Outcomes
PO1	An ability to apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
PO2	Identify, formulate, research and analyze complex engineering problems and reach substantiated conclusions using the principles of mathematics, the natural sciences and the engineering sciences.
PO3	An ability to design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety and of cultural, societal and environmental concerns.
PO4	An ability to conduct investigations of complex problems, considering experimental design, data analysis and interpretation and information synthesis to provide valid conclusions.
PO5	An ability to create, select and apply appropriate techniques, resources and modern engineering and IT tools, including prediction and modeling, to complex engineering activities with an understanding of their limitations
PO6	An ability to apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.
PO7	An ability to understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of and need for sustainable development.
PO8	An ability to apply ethical principles and commit to the professional ethics, responsibilities and the norms of the engineering practice.
PO9	An ability to function effectively as an individual and as a member or leader of diverse teams and in multidisciplinary settings.
PO10	An ability to communicate effectively about complex engineering activities with the engineering community and with society at large. Be able to comprehend and write effective reports, design documentation, make effective presentations and give and receive clear instructions.
PO11	An ability to demonstrate knowledge and understanding of engineering and management principles and apply these to one's work as a team member or a leader to manage projects in multidisciplinary environments.
PO12	An ability to recognize the need for and have the preparation and ability to engage in independent, life-long learning in the broadest context of technological change.